MIL-H-13071D
16 March 1983
SUPERSEDING
MIL-L-13071C
3 April 1969

MILITARY SPECIFICATION

HOUSING, LAMP, RESILIENT MOUNTING,

FOR CONSTRUCTION MACHINERY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a watertight lamp housing with a resilient lamp mounting designed to accommodate a construction machinery type PAR-46 headlamp, floodlamp, or spotlamp.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-L-50	- Lamp, Incandescent, (Sealed Reflector, General Specification for).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6220

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MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-46168	- Coating, Aliphatic, Polyurethane, Chemical Agent Resistant.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging & Supplies and Equipment.
MS17163	- Lamp Incandescent, PAR-46, 12 Volt Floodlamp, Mogul End Prong Base.
MS17165	- Lamp Incandescent, PAR-46, 24 Volt Floodlamp, Mogul End Prong Base.
MS17166	- Lamp Incandescent, PAR-46, 24 Volt Headlamp, Mogul End Prong Base.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

INDUSTRIAL FASTENERS INSTITUTE (IFI)

IFI STANDARDS

(Application for copies should be addressed to Industrial Fasteners Institute, 1517 Terminal Tower, Cleveland, OH 44113.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

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(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The lamp housing (see figure 1) shall consist of a metal housing, mounting bolt, electrical wiring, and connector for a 2-contact-lug-base lamp (see 6.4.1), and a resilient type material or shock absorbing mechanism for holding a PAR-46 sealed-beam lamp. When specified (see 6.2), the lamp housing shall be furnished with one of the following lamps:

MS or Fed. Spec No.	Trade No.	Service
W-L-50	-	-
MS17165	4578	24 volt floodlamp
MS17166	4880	24 volt headlamp
MS17163	4478	12 volt floodlamp

3.2 First article (preproduction model). The contractor shall furnish a lamp housing for examination and testing within the time frame specified (see 6.2) to prove, prior to starting production, that his production methods and choice of design detail will produce lamp housings that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Metals. All metals used in the construction of the lamp housing shall be corrosion-resistant or shall be suitably protected to resist corrosion. The use of dissimilar metals shall be avoided whenever practicable.

3.4 Construction. The lamp housing shall be constructed so that no parts will work loose in service, and replacement of the lamp can be made by personnel without the use of special tools. The lamp housing shall be built to withstand without damage (see 6.4.2) or malfunction the strains, jars, vibrations, and other conditions incident to shipment, storage, installation, and service. The exact shape and design of the lamp housing is optional, provided that all requirements specified herein are met.

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3.5 Operation. The lamp housing, with a lamp installed, shall operate as specified herein without damage or malfunction when tested as specified in 4.5.2.2.

3.5.1 Vibration. The lamp housing shall be capable of operating a lamp for 8 hours, permitting the light beam to shift no more than 2 degrees while being subjected to the vibration amplitude and frequency range specified in 4.5.2.3.

3.5.2 Shock. When tested as specified in 4.5.2.4, the lamp housing shall provide resilient protection to the lamp and shall withstand without damage or malfunction 5 free falls from a height of 6 feet upon a concrete floor.

3.5.3 Temperature conditions. When tested as specified in 4.5.2.5, the lamp housing, with lamp installed, shall operate and shall permit the removal and reinstallation of the lamp without damage or malfunction at any ambient temperature from 125° F to minus 40° F.

3.5.4 Rainfall. When tested as specified in 4.5.2.6, there shall be no evidence of water inside the lamp housing.

3.6 Treatment and painting. Unless otherwise specified (see 6.2), the lamp housing shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, with the top coat conforming to MIL-C-46168, color camouflage forest green.

3.7 Identification marking. The lamp housing shall be identified in accordance with MIL-STD-130.

3.8 Workmanship. The lamp housing shall be fabricated and finished to insure freedom from defects such as burrs, sharp edges, cracks, rust, defects of soldering, welding, brazing, painting, wiring, riveting, or any other defect that could impair its operation or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. Preproduction inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

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4.3 Preproduction inspection.

4.3.1 Examination. The lamp housing shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The lamp housing shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level II.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 6.5 percent defective.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. Tests shall be conducted in the sequence shown. AQL shall be 6.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. Lamp housings shall be examined for the following defects:

101. Absence of any part or component.
102. Components not as specified.
103. Material not as specified.
104. Treatment and painting not as specified.
105. Absence or incorrectness of identification marking.
106. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Prior to tests, a new lamp shall be installed in the lamp housing. Any construction machinery type PAR-46 lamp may be used except that lamps No. 4080, 4480, and 4880 should not be subjected to the vibration and shock tests specified in 4.5.2.3 and 4.5.2.4, respectively.

4.5.2.2 Operation. Connect the lamp housing to a d.c. power source of rated lamp voltage and operate the lamp. Remove, reinstall, and operate the lamp five times. Failure to operate, evidence of electrical arcing, inability to remove and reinstall the lamp, or any evidence of damage to the lamp or the resilient lamp mounting shall constitute failure of this test.

4.5.2.3 Vibration. The lamp housing, with a lamp installed and operating at rated voltage, shall be securely mounted in its normal operating position on a vibration table and vibrated for a continuous period in a vertical plane for 8 hours at a frequency varying uniformly from 10 to 55 cps and return to 10 cps in 1 minute with a total excursion of 0.12 inch. Shift of the light beam more than

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2 degrees between frequency range (10 to 55 cps), loose fittings, broken components, or lamp failure (due to malfunctioning of the lamp housing) shall constitute failure of this test.

4.5.2.4 Shock. The lamp housing, with a lamp installed, shall be mounted in its normal operating position on a 1-inch thick by 12-inch diameter piece of wood cushioned on the bottom by a 1-inch thick by 12-inch diameter piece of honeycomb cardboard. The mounted lamp housing shall be dropped five times from a height of 6 feet onto a concrete floor and in such a manner that the shock shall be imparted to the flat surface of the honeycomb cardboard. After the fifth drop, the lamp shall be connected to a power source. Failure of the lamp to operate (due to malfunctioning of the lamp housing) or permanent damage to the housing or electrical components shall constitute failure of this test.

4.5.2.5 Temperature conditions. The lamp housing, with a lamp installed, shall be placed in a test chamber and the temperature of the chamber shall be set to remain at plus 125° F and minus 40° F each for a period of 12 hours. After the lamp housing has remained at plus 125° F for 12 hours, the lamp shall be removed from and reinstalled in the lamp housing. This test shall be repeated after the housing has been cold soaked for 12 hours at minus 40° F. Inability to remove and reinstall the lamp at 125° F and at minus 40° F, or any evidence of damage to the lamp or to the resilient lamp mounting shall constitute failure of this test.

4.5.2.6 Rainfall. The lamp housing, mounted in its normal operating position with a lamp installed and operating, shall be subjected to a simulated rainfall of 4 inches plus or minus 1 inch per hour for 1 hour, in such a manner as to allow impingement of the rain on the lamp housing (90 degrees) to its horizontal plane. The water distribution shall be such that the rain is dispersed completely over the lamp housing. After exposure to the rain, and with the lamp not operating, sponge or lightly wipe accumulated moisture from the exterior surfaces of the lamp housing, remove the lamp, and visually inspect the lamp housing. Any evidence of water penetration into the lamp housing or failure of the lamp to operate during the entire rainfall shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

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107. Materials, methods, and containers not as specified for level A or B. Each incorrect material, method, or container shall constitute one defect.
108. Contents not blocked in the container to prevent movement or damage in accordance with the referenced document for level A.
109. Box closure and sealing not as specified for level A.
110. Gross weight or size exceeds the limitations specified for level A or B.
111. Preservation and packing not in accordance with the referenced document for commercial.
112. Marking illegible, incorrect, incomplete, or missing for level A, B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each lamp housing shall be preserved in accordance with MIL-P-116, Method III, using a box conforming to PPP-B-636, W6c, style optional. The contents shall be blocked within the box to prevent movement or damage in accordance with MIL-STD-1186. Box closure and sealing shall be as specified for method V in accordance with the appendix to the box specification.

5.1.2 Commercial. Each lamp housing shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The lamp housing, preserved as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional or PPP-B-621, Class 2, style optional, in quantities not to exceed 250 pounds. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification.

5.2.2 Level B. The lamp housing, preserved as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-636, V3c, style optional, or PPP-B-640, Class 2, style optional. The gross weight or size shall not exceed the limitations of the box specification. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.3 Commercial. The lamp housing shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Packages and shipping containers for level A or B shall be marked in accordance with MIL-STD-129.

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pontest corrugated

5.3.2 Commercial. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The intended use of the lamp housing is to provide illumination for construction and industrial machinery normally operated off highways. The lamp housing is designed to accommodate construction machinery type PAR-46 headlamps, floodlamps, and spotlamps and can be mounted on construction machinery having either 6, 12, or 24 volt electrical systems.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Date of DoDISS applicable and exceptions thereto (see 2.1.1).
c. When a lamp is to be furnished and kind of lamp required (see 3.1).
d. Time frame required for submission of preproduction model (see 3.2).
e. Color of paint if other than that specified is required (see 3.6).
f. Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Preproduction model. Any changes or deviations of production lamp housings from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish lamp housings conforming to this specification.

6.4 Definitions.

6.4.1 2-contact lug-base lamp. Commercial designation for a mogul-end prong-base lamp.

6.4.2 Damage. Damage is deformation, loosening, breakage, corrosion or change of fit or any physical condition resulting in noncompliance of the lamp housing with the provisions of this specification.

Custodians:
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Navy - YD
Air Force - 99

User activity:
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Preparing activity:
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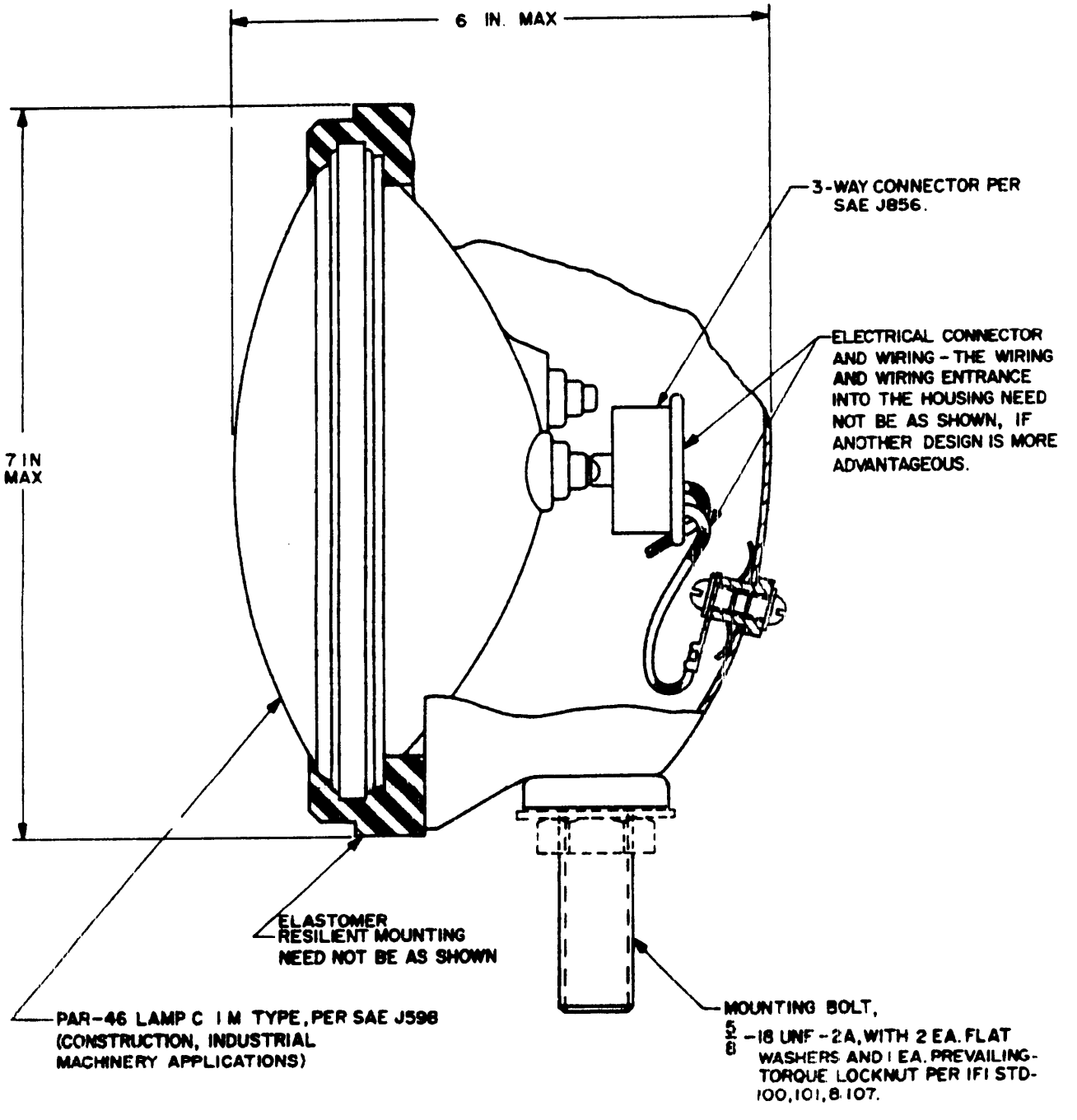


FIGURE 1. Housing, lamp, resilient mounting.

X-250A

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